

IDENTIFICATION

PRODUCT CODE: MAINDEC-X8-DJFPA-A-D
PRODUCT NAME: DEC/X8 MODULE "FPP8-A"
PRODUCT DATE: NOVEMBER, 1976
AUTHOR: WILLIAM J. HEAVEY
MAINTAINER: DIAGNOSTIC ENGINEERING

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1. MODULE DESCRIPTION

"FPP8A" IS A DEC/X8 SOFTWARE MODULE WHICH EXERCISES THE FLOATING POINT PROCESSOR OPTION. THE FPP8A IS A SUBPROCESSOR WITH SINGLE CYCLE DATA BREAK DIRECT MEMORY ACCESS. THIS MODULE OPERATES IN THE FOLLOWING WAY:
ASSIGN A RANDOM BUFFER THEN MODIFY THE FPP8A INSTRUCTION SET AS TO THE MEMORY FIELD AND ADDRESS OF THE BUFFER. THEN LOAD THE "APT" TABLE INTO MEMORY AND LOAD THE FPP BUFFER FIELD AND STARTING ADDRESS POINTER REGISTERS AND START THE FPP8A. WHEN AN INTERRUPT OCCURS (NORMALLY AFTER FIVE SECONDS) CHECK THE FPP ANSWER, INCREMENT THE MODULE COUNTER AND THEN RELEASE THE BUFFER JUST TESTED. THEN ASSIGN A NEW BUFFER AND REPEAT THIS CYCLE. THIS RESULTS IN TESTING THE FPP8A CODE IN ALL EXISTING MEMORY FIELDS.

2. REQUIREMENTS

1. PROCESSORS: PDP-8A, 8/E
2. OPTIONS: FPP8A
3. SPECIAL: NONE

3. RESTRICTIONS

NONE

4. OPERATING INFORMATION

4.1 SPECIAL CONSIDERATIONS

THIS MODULE REQUIRES EXTERNAL BUFFERS.

4.2 BUILDING

1. JOB TYPE: INTERRUPT DRIVEN
2. PRIORITY: NON-CRITICAL
3. JOB SLOT: JF1 OR JF2 ONLY, 4 PAGES REQUIRED.
4. STANDARD DEVICE CODE: 0550

4.3 • INITIALIZING

AFTER "FPP8A" IS PRINTED TYPE THE FOLLOWING PARAMETER:

FOR RANDOM BUFFER USAGE (NORMAL):
FPP8A [0]

FOR A SPECIFIC BUFFER ONLY:
FPP8A [1] [NNNN]
WHERE NNNN IS A LEGAL BUFFER DESIGNATOR

4.4 DEVICE SETUP

NONE

4.5 RUNNING

1.	CNTR:	UPDATED UPON EACH FPP8A INTERRUPT
2.	SR10:	WHEN SET TO A 1, THE BUFFER CURRENTLY ASSIGNED IS RETAINED.
3.	SR11:	NO EFFECT

5. ERROR INFORMATION

ERR8A	INCORRECT EXPONENT
ERR8B	INCORRECT MOST SIGNIFICANT WORD
ERR8C	INCORRECT LEAST SIGNIFICANT WORD
ERR8D	CORRECT EXPONENT
ERR8E	CORRECT MOST SIGNIFICAT WORD
ERR8F	CORRECT LEAST SIGNIFICANT WORD
CODE	BUFFER DESIGNATOR

6. LISTING (ATTACHED)

/DEC/X8 EXTERNAL SYMBOL TABLE "EXTSYM"
/FOR USE IN ASSEMBLING DEC/X8 SOFTWARE MODULES.

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PAUSE

/MAINDEC-X8-DJFPA-A-L "DEC/X8" FPP8A

```

/MODULE TABLE
0200 0200 *200
0201 0000 JOB, 0 /JOB NUMBER
0202 0020 TEXT1, TEXT "FPP8A " /DEVICE NAME
0203 0140
0204 0000
0205 0412 TEXT "DJFPA-A" /MODULE DESIGNATOR
0206 0020
0207 0155
0210 0100
0211 0000 HOMEDF, 0
0212 7402 HLT/COF
0213 5611 JMP I HOMEDF
0214 6202 INTACK, CIF 0
0215 4426 JMS I IHRETP
0216 7777 -1
0217 7777 KILL, -1
0220 7777 KILLED, -1
0221 0000 CNTR, 0
0222 0000 ERROR, 0
0223 3234 OCA ,+11
0224 7604 LAS
0225 0073 AND K4
0226 7440 SZA
0227 3217 OCA KILL
0230 4211 JMS HOMEDF
0231 6002 IOF
0232 6202 CIF 0
0233 4461 JMS I ERHP
0234 0000 0
0235 5622 JMP I ERROR
0236 0000 CODE, 0 /BUFFER AND FIELD POINTER

0237 7772 -6
0240 0000 ERRSA, 0 /INCORRECT EXPONENT
0241 0000 ERRSB, 0 /INCORRECT MSW
0242 0000 ERRSC, 0 /INCORRECT LSW
0243 0015 ERRSD, 0015 /CORRECT EXPONENT
0244 2000 ERRSE, 2000 /CORRECT MSW
0245 0001 ERRSF, 0001 /CORRECT LSW
0246 0000 FPELD, 0 /FPP FIELD POINTER
0247 7777 RENTRY, 7777
0250 0000 FPPW, 0 /FPP INSTRUCTION ADDRESS
0251 0000 FPPX, 0 /FPP APT ADDRESS
0252 0000 FPPY, 0 /FPP IN ADDRESS
0253 0000 FPPZ, 0 /FPP BASE ADDRESS
0254 0000 FPPZA, 0 /FPP ANSWER POINTER

```

```

/FPP-8A ROUTINES
/REINITIALIZE ROUTINE

0255 0000 STFPP, 0
0256 4770 JMS I BUFGEV /ASSIGN A BUFFER
0257 1217 TAD KILL
0260 7450 SNA
0261 5265 JMP ,+4
0262 3220 DCA KILLED
0263 4777 JMS RELEASE
0264 5004 SERVEX
0265 4776 JMS MFLD / FIX THE FPP FIELD
0266 7340 CLA CLL CMA
0267 1251 TAD FPPX /GET THE APT ADDRESS
0270 3017 A113A, DCA AUTO /SAVE IT
0271 4775 JMS CHNFD /CHANGE MEMORY FIELDS
0272 1246 TAD FPELD /GET THE FIELD NUMBER
0273 3417 A113B, DCA I AUTO /FPP-8A P
0274 1250 TAD FPPW /STARTING ADDRESS OF FPP CODE
0275 3417 A113C, DCA I AUTO / P+1
0276 1252 TAD FPPY / GET THE IR POINTER
0277 3417 A113D, DCA I AUTO / P+2
0300 1253 TAD FPPZ /GET THE BASE POINTER
0301 3417 A113E, DCA I AUTO / P+3
0302 3417 A113F, DCA I AUTO / P+4
0303 3417 A113G, DCA I AUTO / P+5
0304 3417 A113H, DCA I AUTO / P+6
0305 3417 A113I, DCA I AUTO / P+7
0306 4211 JMS HOMEDF
0307 1246 TAD FPELD /GET THE FIELD NUMBER AGAIN
0310 0075 AND K7 /MASK TO BITS 9-11
0311 1121 TAD K400 /ADD INTERRUPT ENABLE
0312 6553 DC55B, FPCOM /
0313 6002 IOF
0314 7340 CLA CLL CMA
0315 3247 DCA MENTRY
0316 1251 TAD FPPX /GET THE STARTING ADDRESS OF APT TABLE
0317 6555 DC55C, FPST /STANT FPP-8A
0320 7000 NOP
0321 5655 JMP I STFPP /EXIT

```

```

/FPP-8A INTERRUPT SERVICE ROUTINE
/TEST IF THE CORRECT FPP ANSWER

0322 0000 INT, 0
0323 2247 ISZ MENTRY
0324 7402 HLT /UNEXPECTED INTERRUPT
0325 4211 JMS HOMEDF /RESET DATA FIELD
0326 6552 DC55A, FPICL /RESET FLAG
0327 1374 TAD (FPP1 /GET INTERRUPT SERVICE POINTER
0330 5214 JMP INTACK /EXIT
0331 7300 FPP1, CLA CLL
0332 1254 TAD FPPZA /GET THE APT EXPONENT ADDRESS
0333 3017 A113K, DCA AUTO /SAVE IT
0334 4775 JMS CHNFD /CHANGE MEMORY FIELDS
0335 1417 A113L, TAD I AUTO /GET THE EXPONENT VALUE
0336 3240 DCA ERKSA /SAVE THE EXPONENT
0337 1417 A113M, TAD I AUTO /GET THE MSW
0340 3241 DCA ERKSB /SAVE IT IN BAD
0341 1417 A113N, TAD I AUTO /GET THE LSW
0342 3242 DCA ERKSC /SAVE IT IN BAD
0343 4211 JMS HOMEDF
0344 1240 TAD ERKSA /GET EXPONENT
0345 7041 CIA /NEGATE IT
0346 1243 TAD ERKSD /ADD CORRECT VALUE OF EXPONENT
0347 7640 SZA CLA /ARE THEY EQUAL ?
0350 5362 JMP FPP2 /NO, EXPONENT ERROR
0351 1241 TAD ERKSB /GET MSW
0352 7041 CIA /NEGATE IT
0353 1244 TAD ERKSE /ADD CORRECT VALUE OF MSW
0354 7640 SZA CLA /ARE THEY EQUAL ?
0355 5362 JMP FPP2 /NO, MSW ERROR
0356 1242 TAD ERKSC /GET LSW
0357 7041 CIA /NEGATE IT
0360 1245 TAD ERKSF /ADD CORRECT VALUE OF LSW
0361 7640 SZA CLA /ARE THEY EQUAL ?
0362 4222 FPP2, JMS ERROR /NO, LSW ERROR
0363 4255 JMS STFPP /OK NOW RESTART THE FPP
0364 7300 CLA CLL
0365 2221 ISZ CNTR
0366 7400 H400,
0367 5004 SERVEX

0370 *
0370 0400 BUFGEV, BUFRAN

0374 0331
0375 1106
0376 0424
0377 0416
0400 PAGE

```

/ROUTINE TO ASSIGN A BUFFER OBSERVING SR10
/ENTRY REQUIRES UP=IF

```

0400 0000  BUPMAN, 0
0401 7604  LAS /READ SWITCHES
0402 7012  RTN /MOVE SR 10 IN LINK
0403 7220  CLA CML /
0404 1777* TAD CODE /DESIGNATOR IN AC
0405 7460  SNL SZA /
0406 5214  JMP .+6 /EXIT IF BUFFER ASSIGNED
0407 7640  SZA CLA / AND SR10=1
0410 4216  JMS RELEAS /RELEASE ASSIGNED BUFFER
0411 6002  IOF 0
0412 6202  CIF 0
0413 4460  ASBUFF
0414 3777* DCA CODE
0415 5600  JMP I BUPMAN /EXIT

```

/ROUTINE TO RELEASE A BUFFER

```

0416 0000  RELEAS, 0
0417 1777* TAD CODE /GET BUFFER
0420 6002  IOF /DISABLE INTERRUPTS
0421 6202  CIF 0 /CHANGE TO FIELD 0
0422 4457  RLBUFF /RELEASE THE BUFFER
0423 5616  JMP I RELEAS /EXIT

```

/ROUTINE TO SET UP LOCATION "FPELD" PROPERLY

```

0424 0000  MFLO, 0
0425 7300  CLA CLL
0426 1777* TAD CODE /GET BUFFER DESIGNATOR
0427 7112  RTR CLL /MOVE TO BITS 9-11
0430 7110  RAR CLL
0431 0075  AND K7 /MASK TO BITS 9-11
0432 3776* DCA ERRSA /SAVE THE VALUE
0433 1140  TAD M4 /SET UP A COUNTER
0434 3775* DCA ERRSB / LOCATION
0435 1776* TAD ERRSA /GET VALUE
0436 2775* ISZ ERRSB /FINISHED ?
0437 7410  SKP /NO
0440 5244  JMP .+4 /YES
0441 7106  RTL CLL /MOVE LEFT 3
0442 7104  RAL CLL
0443 5235  JMP .+6 /TRY AGAIN
0444 3774* DCA FPELD /SAVE THE FPP8A FIELD POINTER

```

/ROUTINE TO MODIFY THE FPP INSTRUCTION FIELD BITS
/ AND ADDRESSES

```

0445 1774* TAD FPELD
0446 0075  AND K7 /MASK TO BITS 9-11
0447 3776* DCA ERRSA /SAVE THE VALUE
0450 1324  TAD PNT1 /GET A POINTER
0451 3316  DCA FSAVE1 /SAVE IT
0452 1777* TAD CODE /GET THE BUFFER DESIGNATOR
0453 0131  AND K7600 /MASK TO BITS 0-4
0454 3320  DCA FSAVE3 /SAVE THE VALUE
0455 1322  TAD M23 /SET UP A COUNT
0456 3321  DCA FSAVE4 / LOCATION
0457 1716  FIXITA, TAD I FSAVE1 /GET A VALUE
0460 3317  DCA FSAVE2 /OF THE TABLE
0461 1717  TAD I FSAVE2 /GO INDIRECT
0462 0315  AND K7770 /MASK TO BITS 0-8
0463 1776* TAD ERRSA /ADD "FIELD BITS"
0464 3717  DCA I FSAVE2 /SAVE IT
0465 2317  ISZ FSAVE2 /INCREMENT THE POINTER
0466 1717  TAD I FSAVE2 /GET THE NEXT VALUE
0467 0130  AND K177 /MASK TO BITS 5-11
0470 1320  TAD FSAVE3 /ADD HIGH ORDER ADDRESS
0471 3717  DCA I FSAVE2 /SAVE VALUE
0472 2316  ISZ FSAVE1 /INCREMENT POINTER
0473 2321  ISZ FSAVE4 /FINISHED ALL CHANGES ?
0474 5257  JMP FIXITA /NO, MORE INSTRUCTIONS TO CHANGE
0475 4773* JMS FIXAG /FIX SOME MORE FPP INSTRUCTIONS

```

/ROUTINE TO MOVE 400 LOCATIONS IN THIS MODULE
/ TO THE BUFFER ASSIGNED

```

0476 1323  MOVEIT, TAD PNT2 /GET A POINTER
0477 3316  DCA FSAVE1 /SAVE IT
0500 1772* TAD M400 /SET UP A COUNT
0501 3771* DCA ERRSC / LOCATION
0502 4770* JMS HOMEDF /RESET DATA FIELD
0503 1716  TAD I FSAVE1 /GET A VALUE
0504 4767* JMS CHNFD /CHANGE TO THE BUFFER FIELD
0505 3720  DCA I FSAVE3 /SAVE VALUE IN THE NEW FIELD
0506 4770* JMS HOMEDF /RESET DF=IF
0507 2316  ISZ FSAVE1 /INCREMENT POINTER
0510 2320  ISZ FSAVE3 /INCREMENT POINTER
0511 2771* ISZ ERRSC /FINISHED ?
0512 5302  JMP MOVEA /NO MORE LOCATIONS TO MOVE
0513 4766* JMS MFLOA /SET UP SOME MORE LOCATIONS
0514 5624  JMP I MFLO /EXIT

```

```
0515 7770 K7770, 7770
0516 0000 FSAVE1, 0
0517 0000 FSAVE2, 0
0520 0000 FSAVE3, 0
0521 0000 FSAVE4, 0
0522 7755 M23, -23
      0523 A,
0523 0000 PNT2, FPPRG
0524 0525 PNT1, .+1
0525 0002 TAG0
0526 0005 TAG1
0527 0016 TAG2
0530 0034 TAG3
0531 0042 TAG4
0532 0052 TAG5
0533 0056 TAG6
0534 0060 TAG7
0535 0065 TAG10
0536 0070 TAG11
0537 0712 TAG12
0540 0714 TAG13
0541 0717 TAG14
0542 0723 TAG15
0543 0733 TAG16
0544 0751 TAG17
0545 0031 TAG20
0546 0710 TAG21
0547 1057 TAG22
0566 1064
0567 1106
0570 0211
0571 0242
0572 0366
0573 0760
0574 0246
0575 0241
0576 0240
0577 0236
      0600 PAGE
```

/FPP-8A INSTRUCTION CODE

```
0600 0002 FPPRG, FCLR
0601 0005 STARTF
0602 1011 TAG0, JGE 1
0603 0605 .+2
0604 0000 FEXIT
0605 1021 TAG1, JLE 1
0606 0610 .+2
0607 0000 FEXIT
0610 0212 FLOA 212
0611 0007 JAC
0612 0002 TJAC, FCLR
0613 0006 STARTD
0614 0005 STARTF
0615 0201 TAG2, FLOA 201
0616 1061 JGT 1
0617 0621 .+2
0620 0000 FEXIT
0621 4201 FMUL 201
0622 3201 FDIV 201
0623 0041 FNOP
0624 2201 FSUB 201
0625 6204 FSTA 204
0626 0002 FCLR
0627 0100 LDX 0
0630 0001 1
0631 1101 TAG20, SETX 1
0632 1010 IR
0633 0030 XTA 0
0634 1041 TAG3, JNE 1
0635 0637 .+2
0636 0000 FEXIT
0637 0110 ADDX 0
0640 7777 7777
0641 0030 XTA 0
0642 1001 TAG4, JEQ 1
0643 0645 .+2
0644 0000 FEXIT
0645 0002 FCLR
0646 0003 FNEG
0647 0020 ATX 0
0650 0002 FCLR
0651 0030 XTA 0
0652 1001 TAG5, JEQ 1
0653 0655 .+2
0654 0000 FEXIT
0655 0004 FNORM
0656 1121 TAG6, JSA 1
0657 0663 TJSA
```

```

0660 1001 TAG7, JEQ 1
0661 0667 TJ38
0662 0000 FEXIT
0663 0041 TJ3A, FNOP
0664 0041 FNOP
0665 1031 TAG10, JA 1
0666 0660 ,+6
0667 0213 TJ38, FLDA 213
0670 1071 TAG11, JAL 1
0671 0673 ,+2
0672 0000 FEXIT
0673 0203 FLDA 203
0674 0003 FNEG
0675 3201 FDIV 201
0676 6211 FSTA 211
0677 0204 FLDA 204
0700 5211 FADDM 211
0701 0211 FLDA 211
0702 4201 FMUL 201
0703 1207 FADD 207
0704 2201 FSUB 201
0705 4202 FMUL 202
0706 6204 FSTA 204
0707 0002 FCLR
0710 1111 TAG21, SETB 1
0711 1020 BASE
0712 1131 TAG12, JSR 1
0713 0717 ,+4
0714 1031 TAG13, JA 1
0715 0722 ,+5
0716 0000 FEXIT
0717 1031 TAG14, JA 1
0720 1021 BASE+1
0721 0000 FEXIT
0722 0203 FLDA 203
0723 1051 TAG15, JLT 1
0724 0726 ,+2
0725 0000 FEXIT
0726 0210 FLDA 210
0727 0101 LDX 1
0730 0027 0027
0731 0011 ALN 1
0732 0003 FNEG

```

```

0733 1001 TAG16, JEQ 1
0734 0736 ,+2
0735 0000 FEXIT
0736 0207 FLDA 207
0737 6211 FSTA 211
0740 0202 FLDA 202
0741 7211 FMULM 211
0742 0211 FLDA 211
0743 6205 FSTA 205
0744 0002 FCLR
0745 0204 FLDA 204
0746 3205 FDIV 205
0747 1206 FADD 206
0750 6206 FSTA 206
0751 2171 TAG17, JNX 171
0752 0000 FPPRG
0753 6205 FSTA 205
0754 0002 FCLR
0755 6206 FSTA 206
0756 0205 FLDA 205
0757 0000 FEXIT

```

/ROUTINE TO MODIFY THE FPP INSTRUCTIONS REFERING
/ TO "IR AND BASE" REGISTERS

```

0760 0000 FIXAG, 0
0761 1232 TAD TAG20+1 /GET IR POINTER
0762 1110 TAD K200 /ADD 200
0763 3232 DCA TAG20+1 /SAVE IR POINTER
0764 1311 TAD TAG21+1 /GET BASE POINTER
0765 1110 TAD K200 /ADD 200
0766 3311 DCA TAG21+1 /SAVE BASE POINTER
0767 1320 TAD TAG14+1 /GET BASE+1 POINTER
0770 1110 TAD K200 /ADD 200
0771 3320 DCA TAG14+1 /SAVE BASE+1 POINTER
0772 5760 JMP I FIXAG /EXIT

```

0773 *.
1000 PAGE

```
1000 0000 APT, 0
1001 0000 0
1002 0000 0
1003 0000 0
1004 0000 0
1005 0000 0
1006 0000 0
1007 0000 0
1010 0000 IR, 0
1011 0000 0
1012 0000 0
1013 0000 0
1014 0000 0
1015 0000 0
1016 0000 0
1017 0000 0
1020 0000 BASE, 0
1021 0000 0
1022 0000 0000
1023 0001 0001
1024 2000 2000
1025 0000 0000
1026 7776 7776
1027 0002 0002
1030 0000 0000
1031 0001 0001
1032 5777 5777
1033 7777 7777
1034 0000 000
1035 0000 0000
1036 0000 0000
1037 0000 0000
1040 0000 0000
1041 0000 0000
1042 0000 0000
1043 0000 BASA, 0000
1044 0000 0000
1045 0007 0007
1046 0002 0002
1047 0000 0000
1050 3000 3000
1051 2000 2000
1052 0000 0
1053 0000 0
1054 0000 0
1055 0000 0
1056 0000 0
1057 0001 TAG22, 1
1060 0612 TJAC
1061 0030 0030
1062 3777 3777
1063 7777 7777
```

```
1064 0000 /ROUTINE TO LOAD THE PROPER POINTER TO FPP LOCATIONS
1065 7300 MFLDA, 0
1066 1777 CLA CLL
1067 0131 TAD CODE
1070 3776 DCA FPPH /FPP INSTRUCTION ADDRESS
1071 1776 TAD FPPH
1072 1110 TAD K200
1073 3775 DCA FPPX /FPP APT ADDRESS
1074 1775 TAD FPPX
1075 1076 TAD K10
1076 3774 DCA FPPY /FPP IN ADDRESS
1077 1774 TAD FPPY
1080 1076 TAD K10
1081 3773 DCA FPPZ /FPP BASE ADDRESS
1082 1775 TAD FPPX
1083 1073 TAD K4
1084 3772 DCA FPPZA /FPP EXPONENT ADDRESS-1
1085 5664 JMP I MFLDA /EXIT
```

```
1106 0000 /ROUTINE TO CHANGE OF TO THE ASSIGNED BUFFER FIELD
1107 3317 CHNPD, 0
1110 1777 DCA CHNSA
1111 0105 TAD CODE
1112 1064 AND K70
1113 3314 TAD KCDF
1114 6201 DCA *+1 /CHANGE MEMORY FIELDS
1115 1317 TAD CHNSA
1116 5706 JMP I CHNPD /EXIT
1117 0000 CHNSA, 0
```

```
1120 4454 /INIT ROUTINE TEST FOR INPUT
1121 1371 INIT, CRLF
1122 3324 TAD (TEXT)
1123 4444 DCA INITA
1124 0000 MESSAGE
1125 4455 INITA, 0
1126 4442 SPACE2
1127 5320 ONEOCT
1130 7640 JMP INIT-1
1131 5334 SZA CLA
1132 1370 JMP *+3
1133 5341 TAD (BUFRAN)
1134 4455 JMP INITB
1135 4443 SPACE2
1136 5320 FOROCT
1137 3353 JMP INIT-1
1140 1367 DCA CONBUF
1141 3766 INITB, DCA BUFGEV
1142 5020 INITEX
```

/ROUTINE TO KEEP A CONSTANT BUFFER
/AS ASSIGNED BY THE INPUT PARAM.

1143 0000 BUFCON, 0
1144 1777* TAD CODE
1145 7041 CIA
1146 1333 TAD CONBUF
1147 7650 SNA CLA
1150 5743 JMP I BUFCON
1151 4770* JMS BUFRAN
1152 5344 JMP BUFCON+1

1153 0000 CONBUF, 0

/RUN ROUTINE

1154 3765* RUN, DCA CNTR
1155 3777* DCA CODE
1156 4764* JMS STFPF
1157 7300 CLA CLL
1160 5004 SERVEX

1161 *.
1164 0255
1165 0221
1166 0370
1167 1143
1170 0400
1171 0201
1172 0254
1173 0233
1174 0252
1175 0251
1176 0250
1177 0236
0001 FIELD 1

0000
0100

0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 10001111

0400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0500 11111111 11111111 11111111 11111111 11111111 00000000 00000011 11111111

0600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11100000

1000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1100 11111111 11111111 11111111 11111111 11111111 11111111 10001111 11111111

1200
1300

1400
1500

1600
1700

2000
2100

2200
2300

2400
2500

2600
2700

3000
3100

3200
3300

3400
3500

3600
3700

4000
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

/LOADER CALL

1200	1121	INIT
1201	1154	RUN
1202	0322	INT

/INTERUPT SERVICE

1203	6557	FPIST
1204	0000	0
1205	0001	1
1206	7402	HLT
1207	7402	HLT
1210	7402	HLT

1211	7763	-15
1212	0270	AI13A
1213	0273	AI13B
1214	0275	AI13C
1215	0277	AI13D
1216	0301	AI13E
1217	0302	AI13F
1220	0303	AI13G
1221	0304	AI13H
1222	0305	AI13I
1223	0333	AI13K
1224	0335	AI13L
1225	0337	AI13M
1226	0341	AI13N

1227	7777	-1
1230	0550	0550
1231	7774	-4
1232	0001	1
1233	0326	DC55A
1234	0312	DC55B
1235	0317	DC55C

1236	0000	0
1237	0000	0

/IOT 'S USED

6557 FPIST=6557
6552 FPICL=6552
6553 FPCOM=6553
6555 FPST=6555
6600 FSTA=6600
6602 FCLR=6602
6600 FLUA=6600
6600 FMUL=6600
6600 PDIV=6600
6600 FSUB=6600
6603 FNEG=6603
1000 FADD=1000
2000 JNX=2000
0000 FEXIT=0000
0041 FNUP=0041
5000 FADDH=5000
7000 FMULH=7000
1070 JAL=1070
1110 SETB=1110
1130 JSR=1130
1030 JA=1030
1050 JLT=1050
0010 ALN=0010
1000 JEQ=1000
0100 LDX=0100
1100 SETX=1100
0030 XTA=0030
1040 JNE=1040
0110 ADDX=0110
0020 ATX=0020
0004 FNORM=0004
1120 JSA=1120
0005 STARTF=0005
0006 STARTD=0006
0007 JAC=0007
1020 JLE=1020
1010 JGE=1010
1000 JGT=1000

0000
0100

0200
0300

0400
0500

0600
0700

1000
1100

1200 11111111 11111111 11111111 11111111 00000000 00000000 00000000 00000000
1300 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000

1400
1500

1600
1700

2000
2100

2200
2300

2400
2500

2600
2700

3000
3100

3200
3300

3400
3500

3600
3700

4000
41004200
43004400
45004600
47005000
51005200
53005400
55005600
57006000
61006200
63006400
65006600
67007000
71007200
73007400
75007600
7700

ADDX 0110
A113A 0270
A113B 0273
A113C 0275
A113D 0277
A113E 0301
A113F 0302
A113G 0303
A113H 0304
A113I 0305
A113K 0333
A113L 0335
A113M 0337
A113N 0341
ALN 0010
APT 1000
ASBUFF 4460
ASBUFF 0060
ATX 0020
AUTO 0017
BABA 1043
BASE 1020
BUFCON 1143
BUFGV 0370
BUFRAN 0400
CHNFD 1106
CHNSA 1117
CNTR 0221
CODE 0236
CONBUF 1153
CRLF 4454
CRLFP 0054
DC55A 0326
DC55B 0312
DC55C 0317
ERROR 0222
ERRP 0001
ERRSA 0240
ERRSB 0241
ERRSC 0242
ERRSD 0243
ERRSE 0244
ERRSF 0245
EXINIT 0020
EXSERV 0004
EXTMEM 0161
FADD 1000
FADDM 5000
FCLR 0002
FDIV 3000
FEXIT 0000
FIXAG 0760

FIXITA 0457
FLDA 0000
FMUL 4000
FMULM 7000
FNEG 0003
FNOP 0041
FNORM 0004
FOROCP 0043
FOROCT 4443
FPCOM 0553
FPELO 0246
FPICL 0552
FP18T 0557
PPP1 0331
PPP2 0362
PPPRG 0600
PPPW 0250
PPPX 0251
PPPY 0252
PPPL 0253
PPPLA 0254
PPST 0555
FSAVE1 0516
FSAVE2 0517
FSAVE3 0520
FSAVE4 0521
FSTA 0000
FSUB 2000
HOMEDF 0211
IHRETP 0026
INIT 1121
INITA 1124
INITB 1141
INITEX 3020
INT 0322
INTACK 0214
IOFMSB 0056
IR 1010
JA 1030
JAC 0007
JAL 1070
JEO 1000
JGE 1010
JGT 1060
JLE 1020
JLT 1050
JNE 1040
JNX 2000
JOB 0200
JSA 1120
JSR 1150
K0 0066

K10 0076
K100 0107
K11 0077
K110 0071
K13 0100
K17 0101
K177 0130
K20 0102
K200 0110
K2000 0122
K212 0111
K215 0112
K240 0113
K260 0114
K272 0115
K277 0116
K3 0072
K30 0103
K301 0117
K32 0067
K323 0120
K4 0073
K40 0104
K400 0121
K5 0074
K5200 0123
K540 0124
K5402 0003
K64 0070
K7 0075
K70 0105
K7510 0125
K7520 0126
K7540 0127
K7600 0131
K77 0106
K7735 0132
K7740 0133
K7750 0134
K7760 0135
K7770 0136
K7771 0137
K7773 0137
K7774 0140
K7775 0141
KCDF 0064
KCIF 0005
KCIFUP 0020
KILL 0217
KILLED 0220
KIOF 0004
LOX 0100

L10N 4440
L10NP 0040
M20 0135
M200 0131
M23 0522
M240 0127
M260 0126
M270 0125
M3 0141
M30 0134
M4 0140
M40 0133
M400 0366
M43 0132
M5 0137
M7 0136
MESSAGE 4444
MESSAGEP 0044
MFLD 0434
MFLDA 1064
MOVEA 0502
MOVEIT 0476
MUL26P 0065
ONEOCP 0042
ONEUCT 4442
PNT1 0524
PNT2 0525
PRNT1 4451
PRNT1P 0031
PRNT2 4452
PRNT2P 0052
PRNT4 4453
PRNT4P 0053
RELEASE 0416
RENTY 0247
RLBUFP 4457
RLBUFP 0057
RUN 1154
SERVEX 5004
SETB 1110
SETX 1100
SPACE2 4455
SPACEP 0035
STARTD 0006
STARTP 0005
STFPP 0255
TAG0 0062
TAG1 0065
TAG10 0065
TAG11 0070
TAG12 0712
TAG13 0714

TAB14	0717
TAB15	0723
TAB16	0733
TAB17	0751
TAG2	0016
TAG20	0031
TAG21	0710
TAG22	1057
TAG3	0034
TAG4	0042
TAG5	0052
TAG6	0056
TAG7	0060
TEXT1	0201
TJAC	0012
TJSA	0063
TJSS	0067
TMOOCP	0041
TMOOCT	4441
TYPE	4450
TYPEP	0050
XTA	0030

ERRORS DETECTED: 0

LINKS GENERATED: 42

RUN-TIME: 4 SECONDS

3K CORE USED